

MYSQL PROJECT

TOPIC: ECOMMERCE SALES ANALYSIS 2026 USING MYSQL

SOURCE: KAGGLE

INTRODUCTION

This project performs a structured analysis of an E-Commerce Sales dataset using MySQL. The dataset contains order-level information including order details, customer IDs, product categories, regions, quantities sold, unit prices, discounts, payment methods, delivery times, customer ratings, and revenue.

The objective of this analysis is to uncover meaningful business insights related to sales performance, customer purchasing behaviour, product category trends, regional performance, discount effectiveness, payment preferences, and operational efficiency. The project leverages core and advanced SQL techniques such as filtering, aggregation, joins, Common Table Expressions (CTEs), window functions, subqueries, views, and ranking functions to answer real-world business questions.

TOOL: MySQL Workbench

Submitted By – Sandeep S

Queries

1. What is the total revenue generated?

```
SELECT SUM(revenue) AS total_revenue FROM ecommerce_sales_analytics_5000;
```



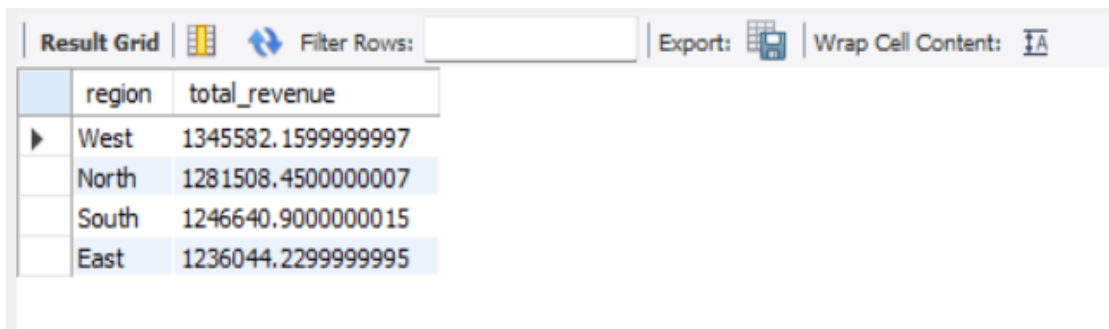
The screenshot shows a MySQL query result grid. The grid has two columns: 'total_revenue' and a value '5109775.740000007'. The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, and buttons for 'Export' and 'Wrap Cell Content'.

	total_revenue
▶	5109775.740000007

Insight: This analysis determines the overall revenue generated from all sales transactions. It provides a baseline measure of the company's financial performance and can be used to compare revenue across products, regions, and time periods.

2. Which region has the highest sales revenue?

```
SELECT region, SUM(revenue) AS total_revenue FROM  
ecommerce_sales_analytics_5000 GROUP BY region ORDER BY total_revenue  
DESC;
```



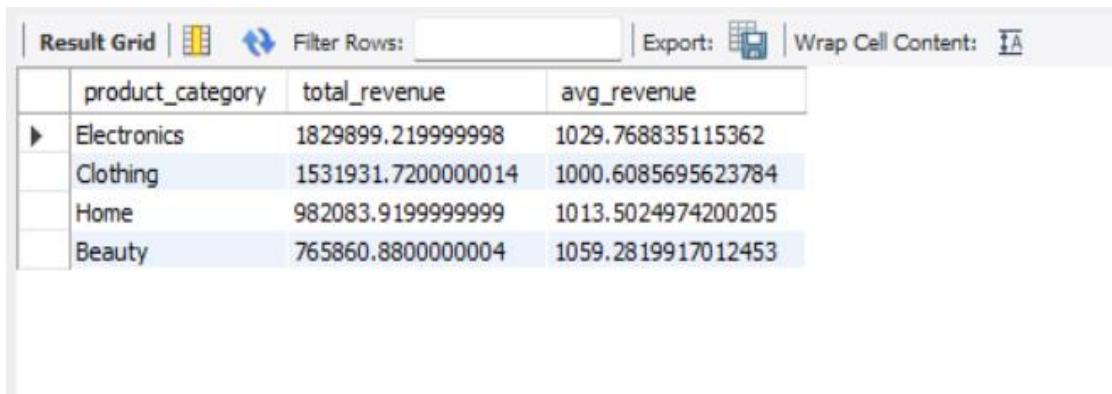
The screenshot shows a MySQL query result grid with two columns: 'region' and 'total_revenue'. The data is sorted in descending order of total revenue. The regions and their corresponding revenues are: West (1345582.1599999997), North (1281508.4500000007), South (1246640.9000000015), and East (1236044.2299999995). The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, and buttons for 'Export' and 'Wrap Cell Content'.

	region	total_revenue
▶	West	1345582.1599999997
	North	1281508.4500000007
	South	1246640.9000000015
	East	1236044.2299999995

Insight: This analysis identifies the region contributing the most sales revenue. The highest-performing region represents a strong market for the business and may provide opportunities for further expansion and targeted marketing.

3. What is the total and average revenue by product category, and which category is the top performer?

```
SELECT product_category, SUM(revenue) AS total_revenue, AVG(revenue) AS  
avg_revenue FROM ecommerce_sales_analytics_5000 GROUP BY product_category  
ORDER BY total_revenue DESC;
```



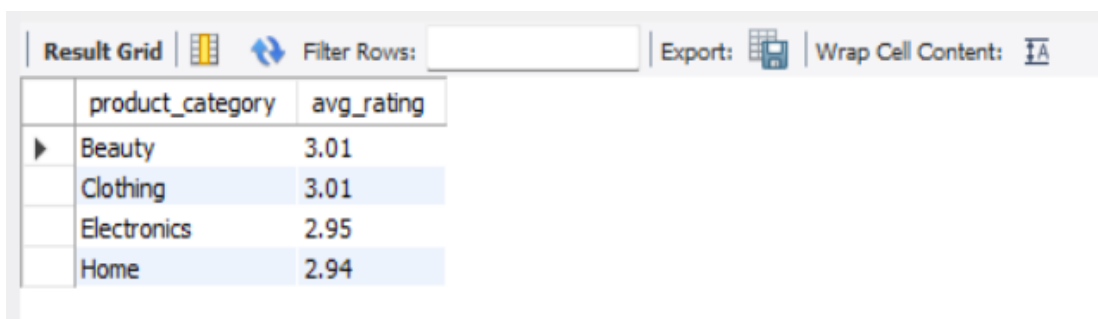
The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a SQL query. The columns are 'product_category', 'total_revenue', and 'avg_revenue'. The rows are sorted by 'total_revenue' in descending order. The categories are Electronics, Clothing, Home, and Beauty.

	product_category	total_revenue	avg_revenue
▶	Electronics	1829899.219999998	1029.768835115362
	Clothing	1531931.7200000014	1000.6085695623784
	Home	982083.9199999999	1013.5024974200205
	Beauty	765860.8800000004	1059.2819917012453

Insight: This analysis compares the total revenue and average revenue generated by each product category to identify the overall top-performing category. A category with the highest total revenue is the strongest contributor to overall sales, while a higher average revenue indicates stronger revenue generation per transaction. This comparison helps the business identify its most profitable categories and make informed decisions about inventory, pricing, and marketing strategies.

4. What is the average customer rating for each product category?

```
SELECT product_category, ROUND(AVG(customer_rating),2) AS avg_rating  
FROM ecommerce_sales_analytics_5000 GROUP BY product_category  
ORDER BY avg_rating DESC;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a SQL query. The columns are 'product_category' and 'avg_rating'. The rows are sorted by 'avg_rating' in descending order. The categories are Beauty, Clothing, Electronics, and Home.

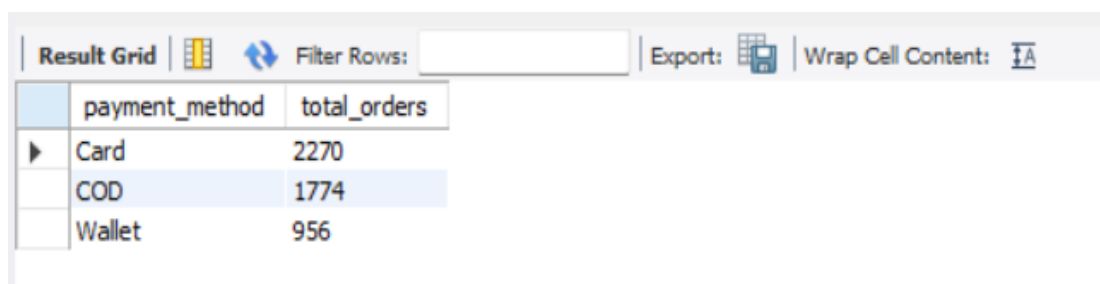
	product_category	avg_rating
▶	Beauty	3.01
	Clothing	3.01
	Electronics	2.95
	Home	2.94

Insight: This analysis measures customer satisfaction across product categories. Categories with higher average ratings indicate better customer satisfaction, while

categories with lower ratings may require further investigation into product quality, service, or customer expectations.

5. Which payment method is used most frequently?

```
SELECT payment_method, COUNT(*) AS total_orders FROM  
ecommerce_sales_analytics_5000 GROUP BY payment_method ORDER BY  
total_orders DESC;
```



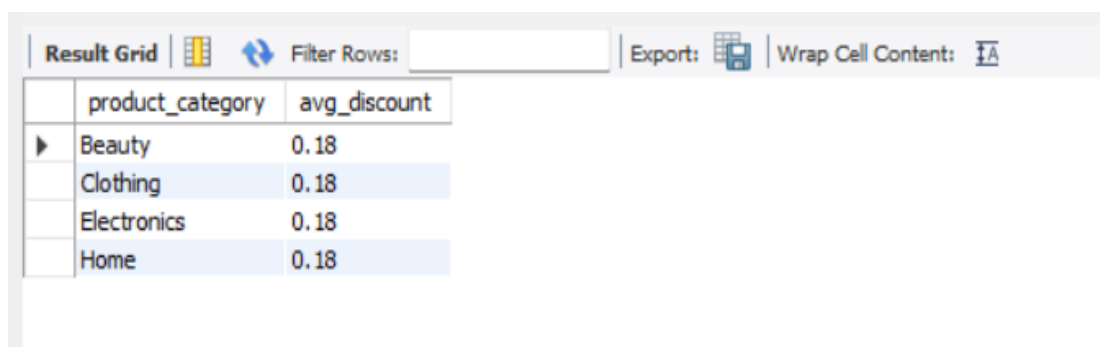
The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with two columns: 'payment_method' and 'total_orders'. The data is as follows:

payment_method	total_orders
Card	2270
COD	1774
Wallet	956

Insight: This analysis identifies the payment method most commonly used by customers. Understanding payment preferences can help the business optimize its payment options and make the purchasing process more convenient for customers.

6. What is the average discount given for each product category?

```
SELECT product_category, ROUND(AVG(discount),2) AS avg_discount  
FROM ecommerce_sales_analytics_5000 GROUP BY product_category;
```



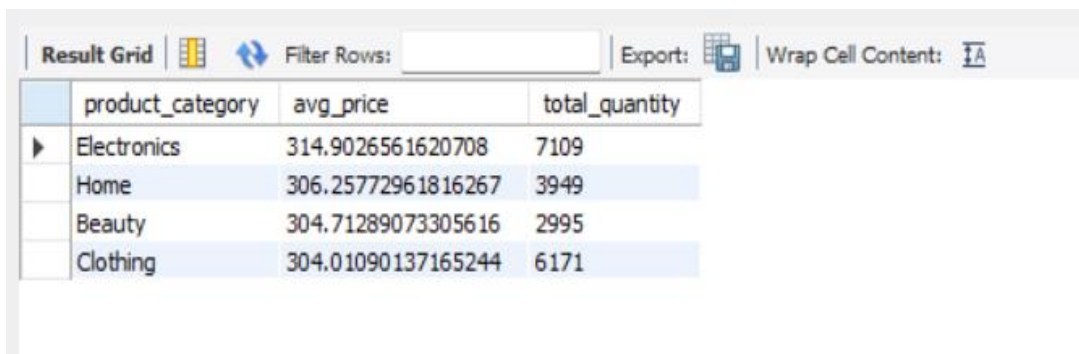
The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with two columns: 'product_category' and 'avg_discount'. The data is as follows:

product_category	avg_discount
Beauty	0.18
Clothing	0.18
Electronics	0.18
Home	0.18

Insight: This analysis compares the average discount offered across product categories. It helps identify which categories depend more heavily on discounts and can be used to evaluate pricing and promotional strategies.

7. Which product_category has the highest average unit_price, and which has the highest sales volume (quantity)?

```
SELECT product_category, AVG(unit_price) AS avg_price, SUM(quantity) AS  
total_quantity FROM ecommerce_sales_analytics_5000 GROUP BY  
product_category ORDER BY avg_price DES;
```



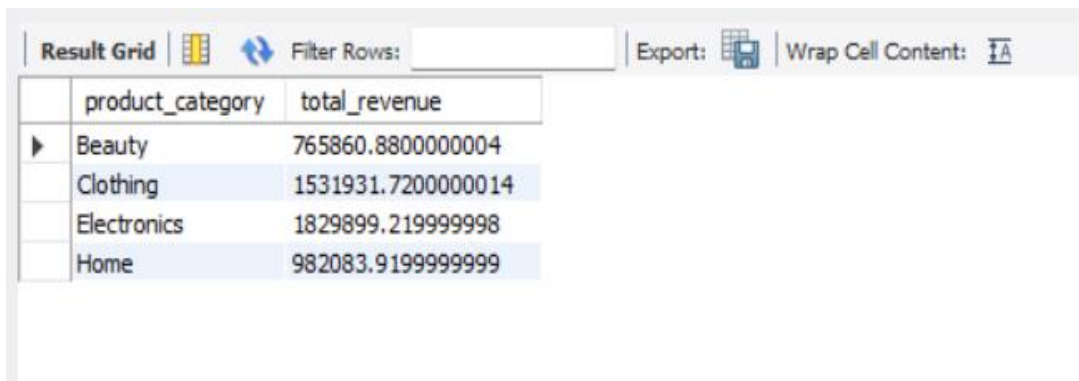
The screenshot shows a database query result grid with the following data:

	product_category	avg_price	total_quantity
▶	Electronics	314.9026561620708	7109
	Home	306.25772961816267	3949
	Beauty	304.71289073305616	2995
	Clothing	304.01090137165244	6171

Insight: This analysis compares product categories from two different perspectives: pricing and demand. A category with the highest average unit price represents higher-priced products, while the category with the highest quantity sold represents the strongest sales volume. Comparing both helps identify whether revenue is driven primarily by premium pricing or high sales volume.

8. Which product categories have generated more than 100,000 in total revenue?

```
SELECT product_category, SUM(revenue) AS total_revenue FROM  
ecommerce_sales_analytics_5000 GROUP BY product_category  
HAVING SUM(revenue) > 100000;
```



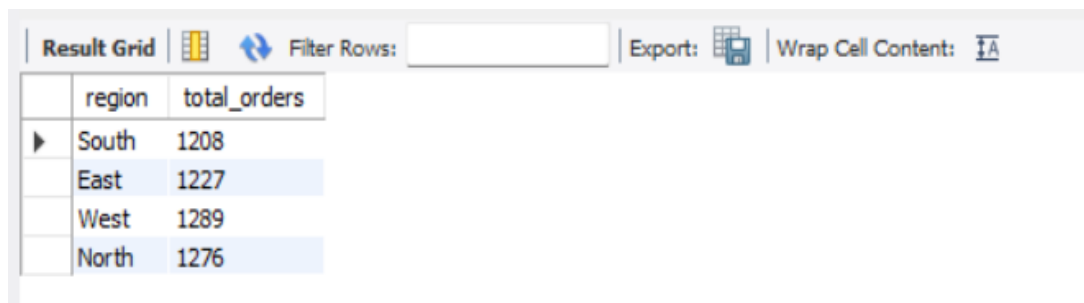
The screenshot shows a database query result grid with the following data:

	product_category	total_revenue
▶	Beauty	765860.8800000004
	Clothing	1531931.7200000014
	Electronics	1829899.2199999998
	Home	982083.9199999999

Insight: This analysis uses a revenue threshold to identify the strongest-performing product categories. Categories generating more than 100,000 in revenue can be considered significant contributors to the business and may deserve greater focus in inventory management, marketing, and sales planning.

9. Which regions have more than 500 orders?

```
SELECT region, COUNT(order_id) AS total_orders FROM
ecommerce_sales_analytics_5000 GROUP BY region HAVING
COUNT(order_id) > 500;
```



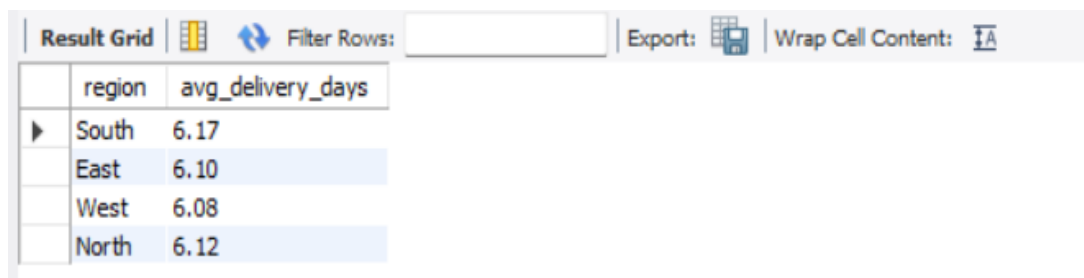
The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a SQL query. The columns are 'region' and 'total_orders'. The rows are: South (1208), East (1227), West (1289), and North (1276). The interface includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' button.

region	total_orders
South	1208
East	1227
West	1289
North	1276

Insight: This analysis identifies regions with a high volume of customer orders. Regions exceeding 500 orders demonstrate strong customer activity and may require greater attention to inventory availability, delivery capacity, and operational resources.

10. Which regions have an average delivery time greater than 5 days?

```
SELECT region, ROUND(AVG(delivery_days), 2) AS avg_delivery_days
FROM ecommerce_sales_analytics_5000 GROUP BY region HAVING
AVG(delivery_days) > 5;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a SQL query. The columns are 'region' and 'avg_delivery_days'. The rows are: South (6.17), East (6.10), West (6.08), and North (6.12). The interface includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' button.

region	avg_delivery_days
South	6.17
East	6.10
West	6.08
North	6.12

Insight: This analysis identifies regions where customers experience relatively long delivery times. Regions with an average delivery time above 5 days may indicate potential logistics or distribution challenges. These regions could be investigated to improve transportation efficiency and customer satisfaction.

11. What are the top 5 highest-revenue orders?

```
SELECT order_id, customer_id, product_category, region, revenue FROM  
ecommerce_sales_analytics_5000 ORDER BY revenue DESC LIMIT 5;
```

order_id	customer_id	product_category	region	revenue
10810	1890	Electronics	South	4119.33
14884	1146	Electronics	South	4083.17
12215	1113	Electronics	South	4055.29
13102	1699	Beauty	East	4005.93
13687	1831	Clothing	North	3928.63

Insight: This identifies the five orders that generated the highest revenue. These orders can be further investigated to determine whether they were driven by high quantities, expensive products, particular categories, or specific regions.

12. What is the average delivery time for each product category?

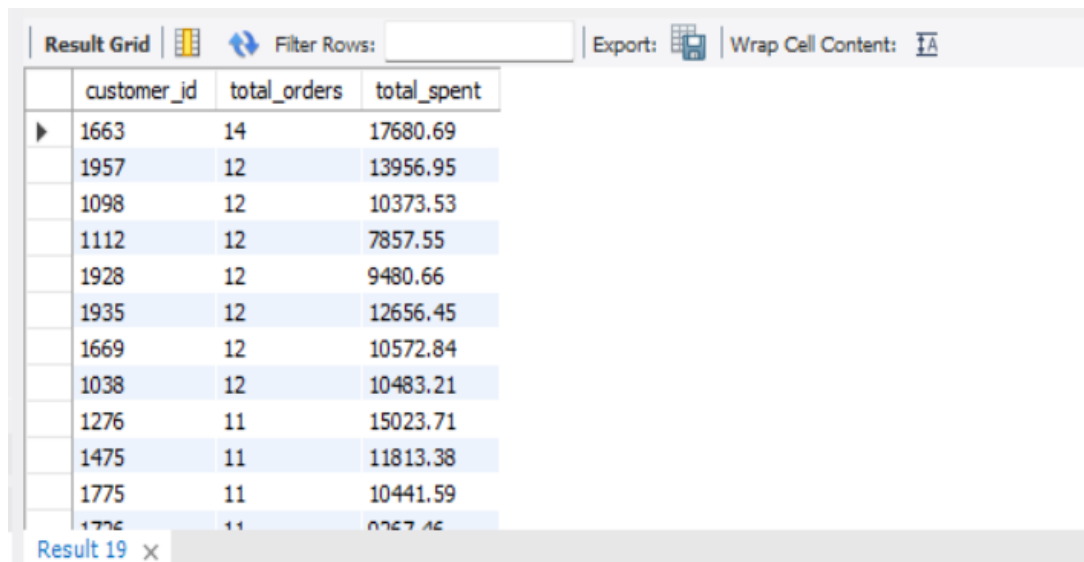
```
SELECT product_category, ROUND(AVG(delivery_days), 2) AS avg_delivery_days  
FROM ecommerce_sales_analytics_5000 GROUP BY product_category  
ORDER BY avg_delivery_days DESC;
```

product_category	avg_delivery_days
Beauty	6.20
Home	6.15
Clothing	6.12
Electronics	6.07

Insight: This comparison highlights which product categories have slower or faster fulfillment. Categories with higher average delivery times may require further investigation to improve logistics and customer satisfaction.

13. Which customers have placed more than 10 orders?

```
SELECT customer_id, COUNT(order_id) AS total_orders, ROUND(SUM(revenue),  
2) AS total_spent FROM ecommerce_sales_analytics_5000 GROUP BY  
customer_id HAVING COUNT(order_id) > 10 ORDER BY total_orders DESC;
```



The screenshot shows a MySQL query result grid with the following data:

	customer_id	total_orders	total_spent
▶	1663	14	17680.69
	1957	12	13956.95
	1098	12	10373.53
	1112	12	7857.55
	1928	12	9480.66
	1935	12	12656.45
	1669	12	10572.84
	1038	12	10483.21
	1276	11	15023.71
	1475	11	11813.38
	1775	11	10441.59
	1776	11	10767.46

Result 19 x

Insight: This identifies high-frequency repeat customers who have placed more than 10 orders. These customers are valuable because of their strong purchasing activity and may be suitable for loyalty programs, personalized offers, or retention strategies.

CONCLUSION

The Sales Data Analysis project successfully demonstrated the use of MySQL for analysing sales, customer, product, regional, and transaction data to generate meaningful business insights. Through the application of SQL concepts such as filtering, aggregation, GROUP BY, HAVING, subqueries, and sorting, the project transformed raw sales records into actionable information.

The analysis provided valuable insights into overall revenue generation, product category performance, regional sales, customer purchasing behaviour, payment method preferences, monthly sales trends, discount patterns, and delivery performance. The analysis also identified top-performing product categories, high-revenue orders, high-value customers, and frequent repeat customers, helping evaluate sales performance and customer behaviour.

Additionally, the project highlighted product categories with different revenue and delivery patterns and identified regions with significant order volumes and longer delivery times. These findings can help businesses improve pricing strategies, customer retention, inventory planning, marketing activities, and logistics operations. Overall, this project strengthened practical SQL and data analysis skills and demonstrated how sales data can be used to support data-driven decision-making and improve overall business performance.